What	Linear Algebra	Fourier Series
Vector Space	$\mathbb{R}^n$	Piecewise smooth 2π-periodic functions on $\mathbb{R}$
Inner Product	$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$	$\langle f(t), g(t) \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) g(t) dt$
Orthonormal Basis $\left(\mathbb{R}^3\right)$	$\{(1,0,0),(0,1,0),(0,0,1)\}$	$\{1, \cos nt, \sin nt\}$
A Vector in the Basis	$\mathbf{x} = a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + \cdots + a_n \mathbf{e}_n$	$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt$
Coefficients are Projections	$a_i = \langle \mathbf{x}, \mathbf{e}_i \rangle$	$a_0 = \langle f(t), 1 \rangle$ $a_m = \langle f(t), \cos mt \rangle$ $b_m = \langle f(t), \sin mt \rangle$

Table 1: Vector Spaces: Linear Algebra vs. Fourier Series

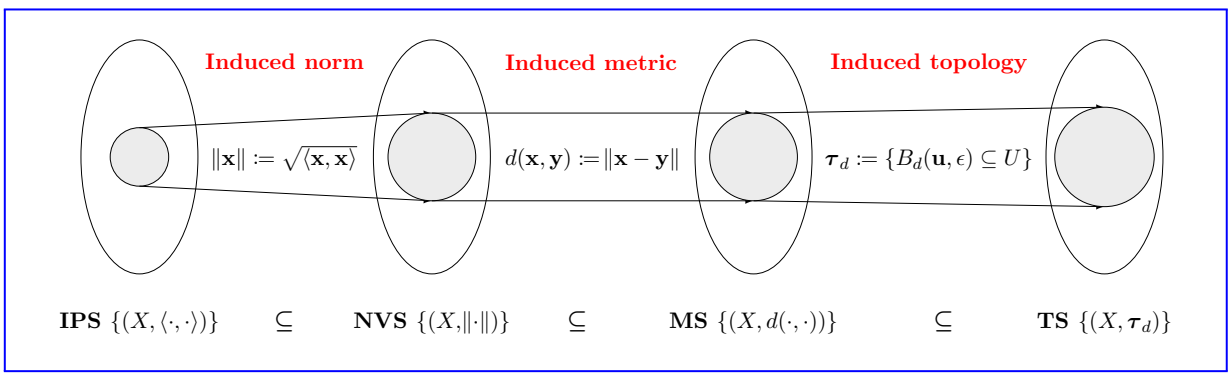

Figure 1: Induced Norm, Metric, and Topology

IPS	$--$	Inner Product Spaces	$(X, \langle \mathbf{x}, \mathbf{y} \rangle)$
NVS	$--$	Normed Vector Spaces	$(X, \|\mathbf{x}\|) = (X, \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle})$
MS	$--$	Metric Spaces	$(X, d(\mathbf{x}, \mathbf{y})) = (X, \|\mathbf{x} - \mathbf{y}\|) = (X, \sqrt{\langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y} \rangle})$
TS	$--$	Topological Spaces	$(X, \boldsymbol{\tau}_d)$, where $\boldsymbol{\tau}_d = \{\text{open subsets of } (X, d)\}$

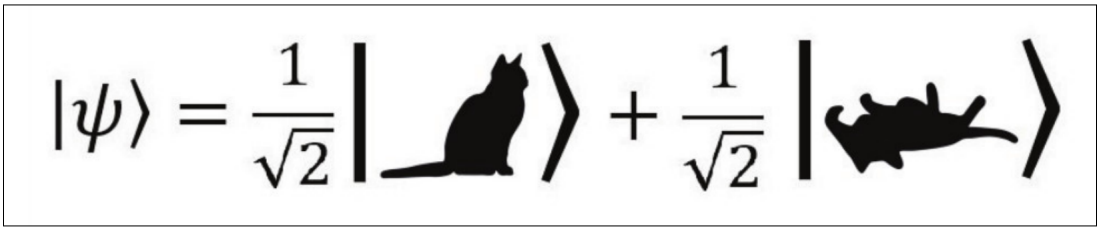

Figure 2: Quantum superposition and Schrödinger's cat

$n=\langle\ ,\ \rangle$
$G(\ )$
$\ \ \ \

0.6249999121401877

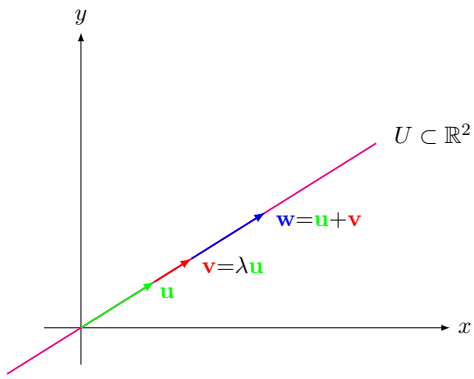

Figure 5: The line U is a linear subspace of $\mathbb{R}^2$

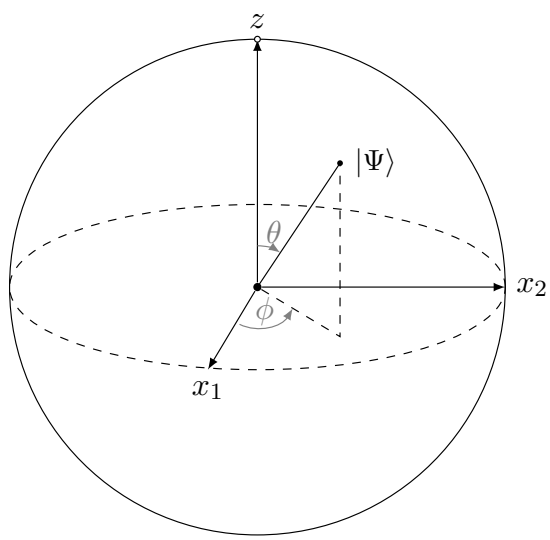

Figure 6: The Bloch Sphere

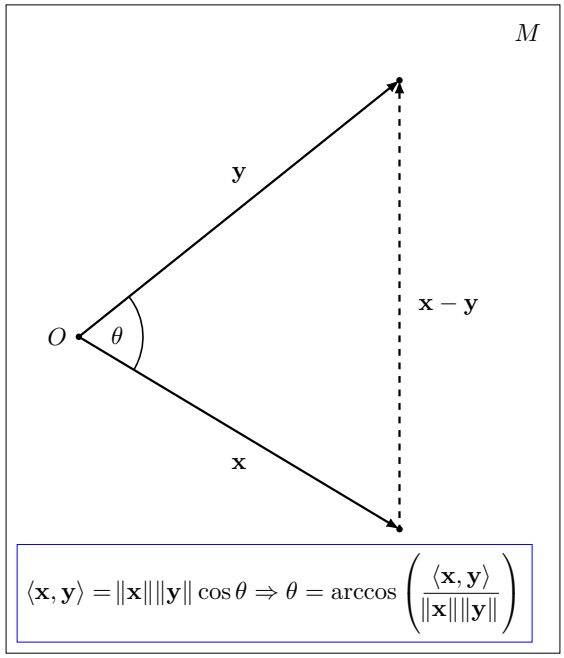

Figure 7: Geometric interpretation of $\langle \mathbf{x}, \mathbf{y} \rangle$

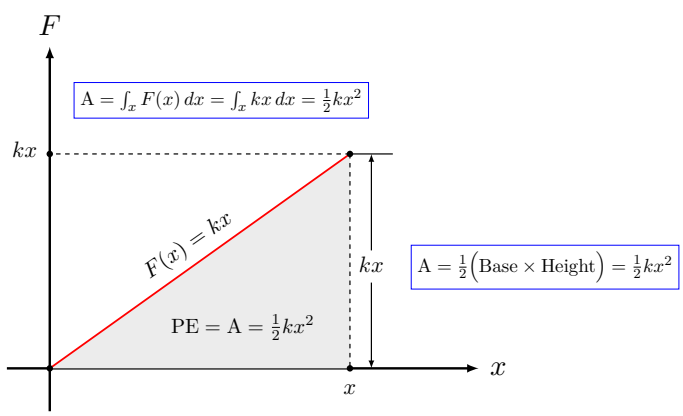

Figure 8: Potential Energy of a Simple Harmonic Oscillator

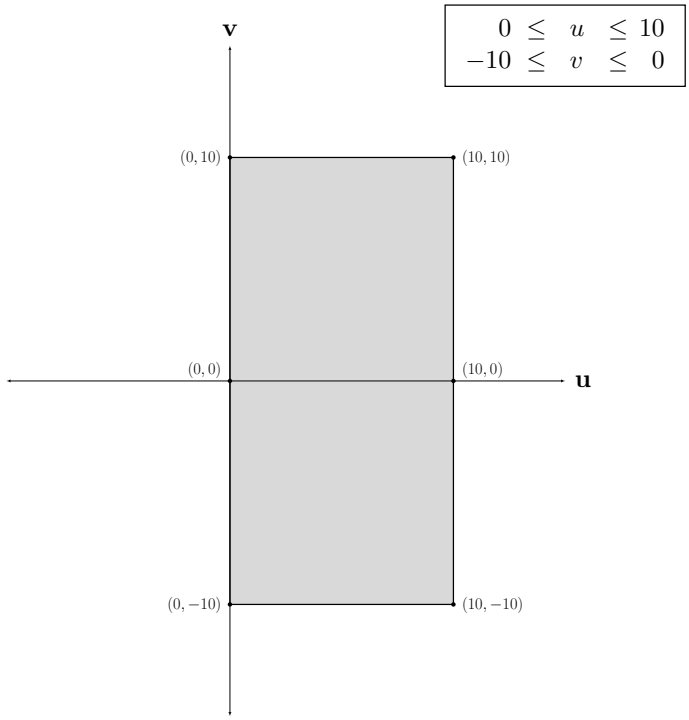

Figure 9: The transformed parallelogram is a rectangle in the uv-plane

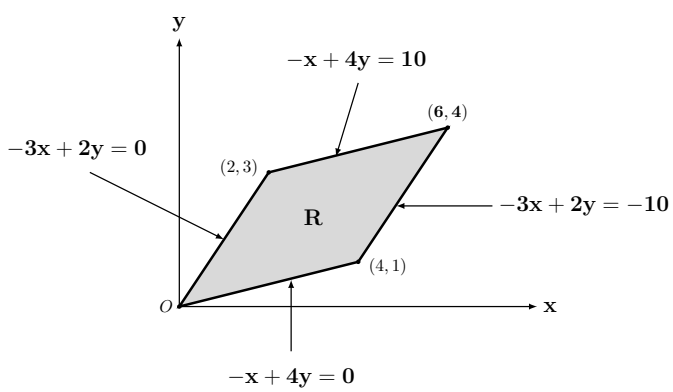

Figure 10: Equations of the Sides of the Parallelogram

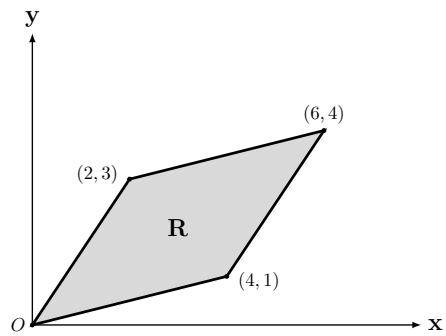

Figure 11: Parallelogram in the xy-plane

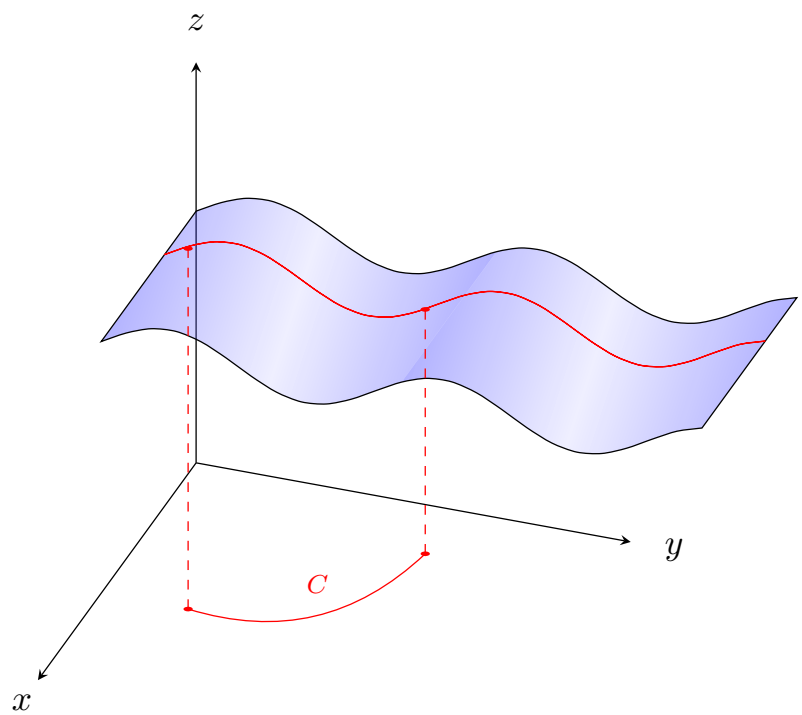

Figure 12: 3D

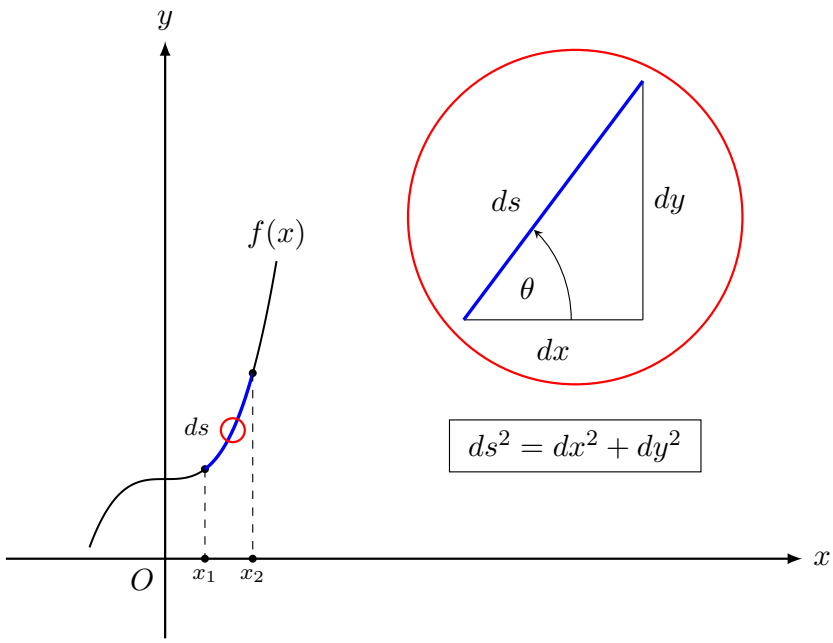

Figure 13: $f(x)$, ds and the Pythagorean Theorem

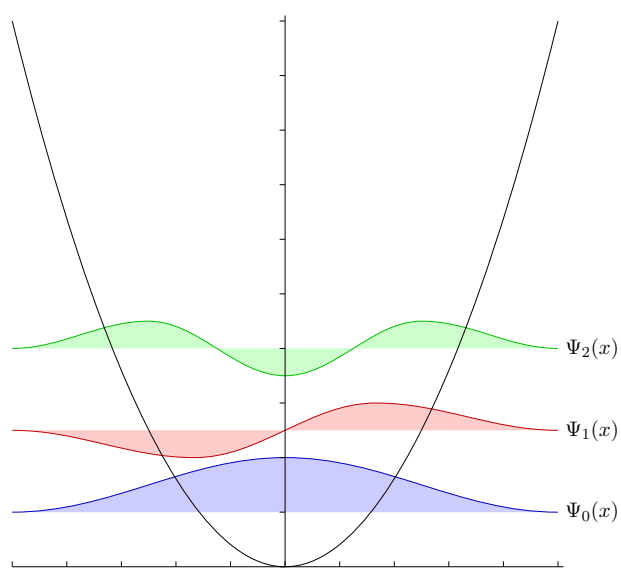

Figure 14: Wave Functions

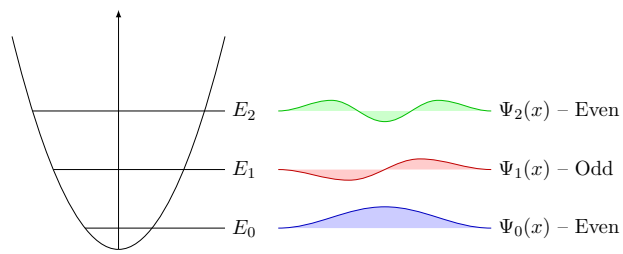

Figure 15: Wave Functions

9^2	$=$	81	$\longrightarrow$	$8+1$	$=$	9
45^2	$=$	2025	$\longrightarrow$	$20+25$	$=$	45
703^2	$=$	494209	$\longrightarrow$	$494+209$	$=$	703
7777^2	$=$	60481729	$\longrightarrow$	$6048+1729$	$=$	7777
857143^2	$=$	734694122449	$\longrightarrow$	$734694+122449$	$=$	857143

Figure 16: A Few Example Kaprekar Numbers

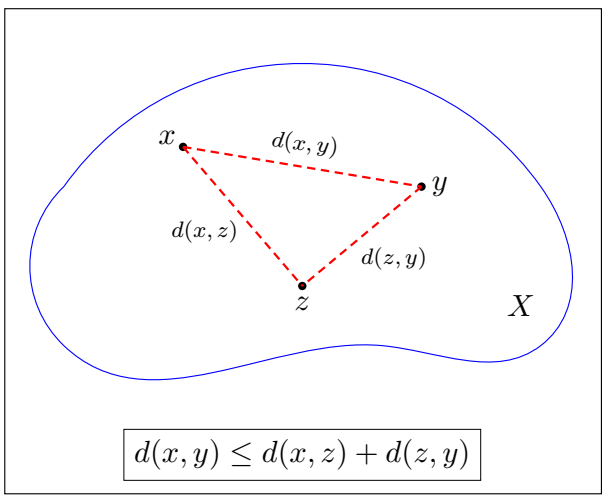

Figure 17: The Triangle Inequality

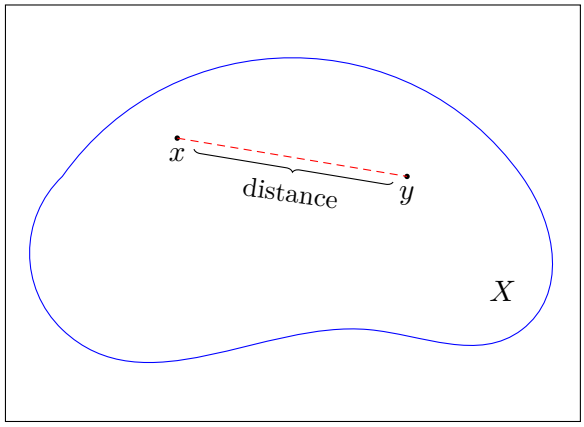

Figure 18: X, x and y

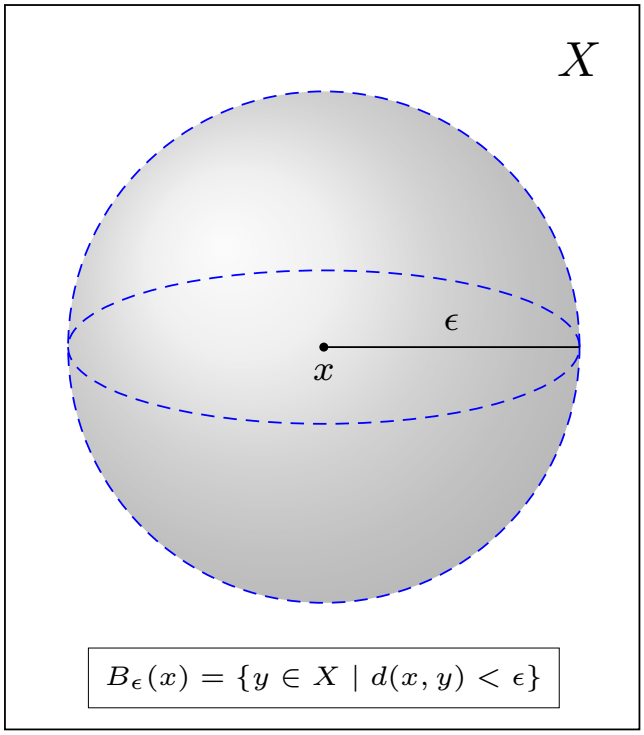

Figure 19: $B_\epsilon(x)$ is an open epsilon ball centered at x with radius ϵ

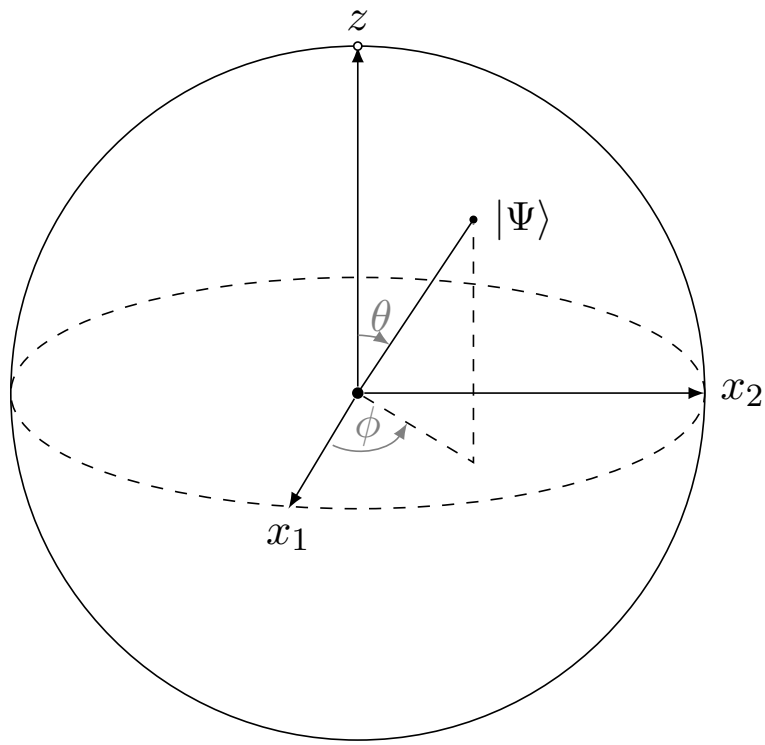

Figure 20: The Bloch Sphere

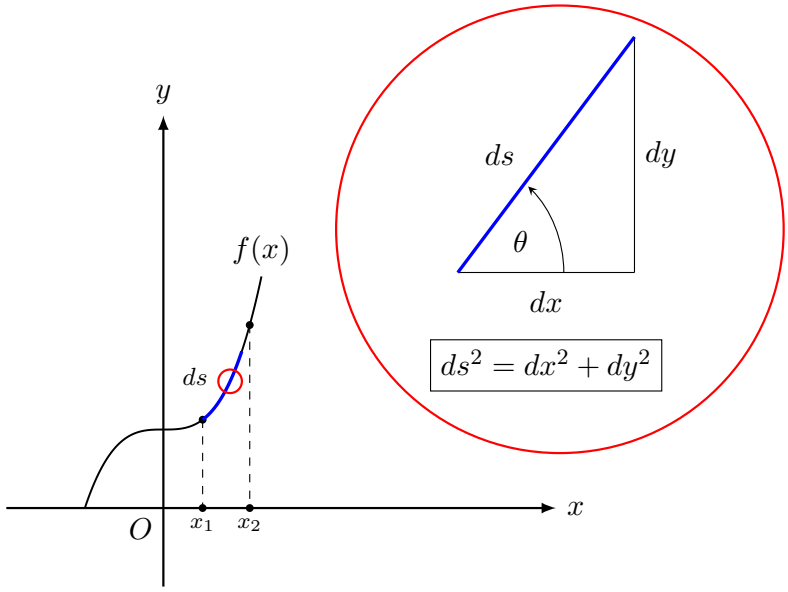

Figure 21: $f(x)$, ds and the Pythagorean Theorem

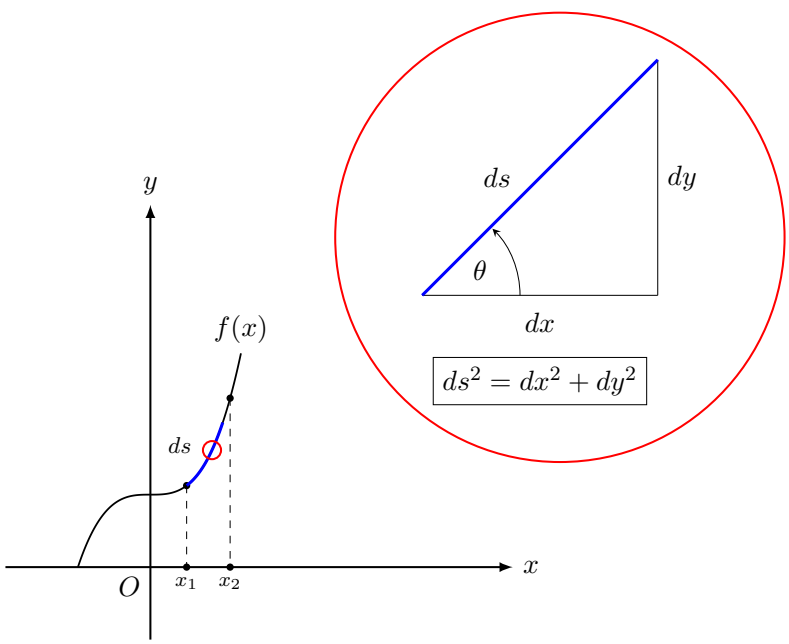

Figure 22: $f(x)$, ds and the Pythagorean Theorem